

P3↔P2 Directed Selection Inheritance

Note #682 — Series CC, Phase CC.2 (CC.16)

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This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, in explicit inheritance-edge form, that Paper 3's DNA absorption governance (D2.11) is directed selection at inter-Self scope — the same mechanism Paper 2 establishes at intra-Self scope, with inter-Self source content as the only addition.

Abstract

Paper 2 establishes directed selection as the primary governance-controlled DNA evolution mechanism: a governed process in which an LLM-mediated proposing substrate surfaces candidates from the action layer, governance reviews them, and governance-authorized changes execute through a directed selection event with version records and authorization attribution. Paper 3's DNA absorption governance (D2.11) is directed selection applied at inter-Self scope. The five steps of the absorption protocol map one-to-one onto the directed selection process: candidate identification (proposing), governance review (review), absorption decision (authorization), directed selection event (execution), and absorption record (versioning). The mechanism is identical; the source content expands to include Full Aspect Integration (FAI)-origin DNA from another Self's contributed aspects. Step 4 of D2.11 executes a directed selection event literally — making the inheritance an identity, not an analogy. Adversarial claims that DNA absorption constitutes a novel inter-organizational governance mechanism must address that Step 4 is precisely the directed selection event Paper 2 establishes. This is the seventeenth note in the Series CC cross-derivation series and the fifth to document the "same mechanism, new sources" extension pattern that characterizes Paper 3's systematic construction on Paper 2's foundations.

1. The Directed Selection Commitment at Paper 2 Scope (B1.14, B1.15)

Paper 2 commits to three evolution mechanisms in productive tension: instinct evolution (undirected, through LLM version changes and infrastructure), DNA evolution (directed selection under human governance), and action-feedback evolution (operationally-driven proposals feeding into governed DNA refinement). Of the three, directed selection is the canonical governance-controlled form.

Paper 2 §7 is explicit that direction here means goal-defined-by-human-governance — not goal-defined-by-internal-system-state, fitness function, or emergent bias. Selection criteria are authored and maintained by human governance, are themselves substrate content under the same authority architecture, and are modifiable. This is what distinguishes directed selection architecturally from biological selection and from autonomous optimization loops: the governing criteria are external to the mechanism that applies them and remain under human authority throughout.

At intra-Self scope, directed selection operates through a governed three-step process. First, a proposing substrate — an LLM-mediated component operating under governance-authored orchestration rules — reads the action layer and surfaces improvement proposals for home governance to review. The proposing substrate is itself substrate content: its outputs are proposals, not automatic changes. Second, governance review: human governance practitioners examine the proposals, applying governance-defined success criteria, and decide which to authorize. Third, a directed selection event: governance-authorized changes execute as governed modifications to the DNA layer, with version records and authorization attribution created at the time of the change.

The authority architecture for directed selection (B1.15) specifies who can propose, who can authorize, what verification regime applies, and what reversion paths exist. These specifications are themselves substrate content — recursively governed under the same authority architecture they specify. DNA changes may be drafted by LLMs operating under human direction, with humans holding authority over the orchestration rules under which LLMs propose and over the acceptance decisions.

At intra-Self scope, the source content for directed selection candidates is the Self's own action layer: accumulated records of what the Self has executed, what patterns emerged, and what operational evidence warrants consideration for DNA refinement. The proposing substrate scans this intra-Self content and generates proposals against it.

2. DNA Absorption Governance at Inter-Self Scope (D2.11)

Paper 3 extends the evolution architecture to the inter-Self perimeter through Full Aspect Integration (FAI). When a Self contributes aspects to a shared substrate during a FAI event, DNA-layer content — orchestration substrates, schemas, behavior rules — from another Self's contributed aspects becomes available to the receiving Self's home evolution

machinery after the event dissolves.

Where governance authorizes it, orchestration patterns and rules from another Self's contributed aspects can be selectively absorbed into the receiving Self's DNA layer. Paper 3 specifies that this absorption operates through Paper 2's DNA evolution mechanism — directed selection under home governance. The receiving Self's authority structure governs which DNA content from the shared substrate is a candidate for absorption, which is approved, and which is declined or deferred.

Paper 3's DNA absorption governance specifies a five-step protocol for this process (D2.11):

Step 1 — Candidate Identification: Home governance identifies FAI-origin DNA content as candidates for absorption consideration. This is the inter-Self equivalent of the proposing substrate surfacing action-feedback candidates from the Self's own records — the source has shifted from intra-Self action-layer to inter-Self FAI-origin DNA, but the identifying function is the same.

Step 2 — Governance Review: Home governance reviews the identified candidates, applying its own governance-defined success criteria to the inter-Self source content.

Step 3 — Absorption Decision: Governance makes explicit absorb, decline, or defer decisions for each candidate. The decision is governance's, not the mechanism's: no absorption is automatic, and decline or deferral are architecturally equal outcomes to absorption.

Step 4 — Directed Selection Event: Approved absorption is executed through a directed selection event, with governance authorization recorded at the time of execution.

Step 5 — Absorption Record: The absorption is documented with FAI-origin provenance, recording both the DNA change and its inter-Self source for downstream auditability.

3. The Five-Step Mapping: Absorption IS Directed Selection

The inheritance claim is an identity, not an analogy. The five absorption steps map one-to-one onto the directed selection process Paper 2 establishes:

D2.11 Absorption Step	Paper 2 Directed Selection Equivalent
Step 1: Candidate Identification	Proposing substrate surfaces candidates
Step 2: Governance Review	Governance reviews proposals
Step 3: Absorption Decision	Governance authorizes or declines
Step 4: Directed Selection Event	Directed selection event executes the change

Step 5: Absorption Record	Version record with authorization attribution
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The mapping is not structural correspondence that requires interpretive bridging — it is a point-by-point identity. Step 4 does not resemble a directed selection event; it is one, executed through Paper 2's mechanism, producing precisely the version records and authorization attribution Paper 2 specifies. The protocol label is different; the mechanism is the same.

The critical inheritance point is Step 4's explicitness. Paper 3 does not describe Step 4 as an absorption-specific novelty; it describes it as execution through a directed selection event — the named mechanism Paper 2 establishes as the canonical governance-controlled DNA change form. The use of the same named mechanism is not coincidental. It is the inheritance claim: absorption at inter-Self scope executes through directed selection at intra-Self scope. The same machinery handles both.

The sole architectural addition is the origin of source content. At intra-Self scope, the proposing substrate surfaces candidates from the Self's own action layer. At inter-Self scope, candidates also come from FAI-origin DNA: another Self's contributed aspects, carried into the shared substrate during a FAI event and available for home governance to consider. The identifying function changes its input domain; the governed process that follows is identical across all five steps.

4. What Is Preserved; What Is Extended

Preserved without modification:

Directed selection as the governance-controlled DNA change mechanism. Governance review and authorization required for every DNA change, regardless of whether the candidate originated intra-Self or inter-Self. Directed selection event execution with version records and authorization attribution at each change. The authority architecture specifying who proposes, who authorizes, what verification regime applies, and what reversion paths exist. The recursive governability property: selection criteria are themselves substrate content under the same authority architecture they are applied within, and that property carries intact through the absorption protocol. The asymmetry of labor and authority: LLMs may draft, identify, and prepare absorption candidates under human direction; humans hold authority over the acceptance decisions.

Extended at inter-Self scope:

The source content for directed selection candidates expands to include FAI-origin DNA from another Self's contributed aspects — content that originated outside the receiving Self's home governance perimeter. The receiving Self's home governance retains full authority over the absorption decision, the verification regime, and the reversion path. Asymmetric ingestion holds: what one Self absorbs from a FAI event is governed within that

Self's home authority and may differ entirely from what the other participating Self absorbs from the same event. The extension is input-side only. Nothing in the absorption protocol requires new governance machinery, new authorization architecture, or new versioning primitives.

The FAI-origin provenance record in Step 5 is the one piece of the absorption protocol that has no direct intra-Self equivalent. At intra-Self scope, the source of a directed selection candidate is always the Self's own action layer; provenance attribution is intra-perimeter. At inter-Self scope, the source is another Self's DNA, and the absorption record must carry that inter-Self origin so that downstream governance can audit what came from where. This is an extension of the record-keeping commitment, not a new mechanism: the same versioning structure carries an additional provenance field.

5. The Prior-Art Inheritance Claim

Paper 3's DNA absorption governance IS directed selection at inter-Self scope. The five-step absorption protocol is the directed selection protocol with inter-Self source content. The identity is established by D2.11's own specification: Step 4 executes a directed selection event — the term Paper 2 uses for the mechanism — using Paper 2's machinery. No interpretive argument is required; the protocol names the mechanism it uses.

Adversarial claims that inter-organizational AI governance learning through DNA absorption constitutes a novel mechanism not previously established in the trilogy must address one concrete obstacle: Step 4 of the D2.11 absorption protocol is a directed selection event as Paper 2 specifies. The mechanism is not novel. The source content adds an inter-Self origin. Adding new sources to an established mechanism does not constitute a new mechanism — it extends the mechanism's input domain while the governed process structure remains intact.

This note is the seventeenth in the Series CC cross-derivation series and the fifth to document the "same mechanism, new sources" pattern across CC.1x and CC.2x notes. Across these five inheritance edges, Paper 3's architecture has extended Paper 2's action-feedback evolution, mating operation, conflict handling, evolution-feed routing, and now directed selection to the inter-Self scope by expanding their input sources while preserving their governed process structures. The accumulation of five such edges, each formalizing the same structural move across distinct mechanisms, establishes that the pattern is systematic. Paper 3 is not a collection of independent additions to Paper 2; it is a coherent architectural extension in which inter-Self scope is achieved by applying Paper 2's mechanisms to inter-Self source content under the same governance architecture that governs intra-Self operation.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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